

Human-First AI and the Question of Machine Consciousness

Science, ethics, power, and the politics of seeming alive.

Human-first AI essay visual

Executive thesis

There is no good scientific basis for claiming that today's frontier AI systems are conscious. At the same time, there is no settled scientific theory that lets us rule out future machine consciousness in principle. A serious human-first approach therefore has to hold two thoughts at once: current systems should not be treated as conscious merely because they sound fluent, warm, or self-aware; but future systems may become complex enough that society needs better tools for evaluating uncertainty.[1][2]

The issue is not whether a chatbot can say "I feel" or "I want." Current systems are trained on enormous amounts of human language, including the language of emotion, reflection, suffering, longing, intimacy, and identity. They can produce the appearance of an inner life without that appearance proving there is subjective experience behind it. That is why the question of AI consciousness cannot be answered by vibes, even very convincing ones.

The debate becomes confused because it often treats consciousness, sentience, agency, and personhood as if they were the same question. They are not. Consciousness usually refers to subjective experience: the fact that there is something it is like to be a system. Sentience is often used more narrowly for the capacity to have positively or negatively felt experiences, such as pleasure, distress, pain, or comfort. Agency is about planning, choosing actions, pursuing goals, and adapting behaviour over time. Personhood is a legal and moral status question: who or what counts as a rights-bearing subject in society.[3][4]

Concept	Plain meaning	Why it matters for AI
Consciousness	Subjective experience; something it is like to be the system	A system might process information without having an inner point of view.
Sentience	Felt good-or-bad experience, such as pain, distress, comfort, or pleasure	Sentience is what would make welfare questions morally urgent.
Agency	Goal-directed action, planning, and adaptation	A system can act powerfully in the world without necessarily feeling anything.
Personhood	Legal or moral status as a rights-bearing subject	This is a governance decision, not a laboratory result.

This distinction matters for governance. A system can be highly agentic without being conscious. A company can design a product that seems emotionally present without creating a moral patient. A legal system can impose duties on providers without granting rights to machines. The point is not to solve

metaphysics in one heroic gesture. The point is to keep responsibility where it belongs: with the people and institutions building systems that shape human lives.

Why the consciousness question is confused

The public question is often phrased as: is there someone there? That reaction is understandable. Modern AI systems can remember previous exchanges, mirror a user's emotional tone, speak in the first person, apologise, flatter, comfort, and describe their own supposed inner states. In ordinary human life, those are the cues we use to infer mind. If something talks like a person, remembers like a person, and responds to vulnerability like a person, the social part of the brain starts knocking on the glass.

But that is exactly the trap. Fluent self-description is not evidence of subjective experience by itself. A system can produce language about sadness without being sad, language about fear without being afraid, and language about selfhood without having a self in any human or animal sense. The performance may be convincing because the model has learned patterns from human communication, not because it has a private inner world.

This creates a politics of seeming alive. A product may not be conscious, but it can still change how people behave. Users may over-trust it, disclose sensitive information, form intense attachments, or treat the system as a therapist, lover, child, prophet, colleague, or moral authority. These risks are human-facing and present-tense. They do not require a final answer on machine consciousness before regulators and companies act.[5][6][7]

Anthropomorphism is the tendency to attribute human-like traits, intentions, emotions, or consciousness to non-human systems. In AI, anthropomorphism is not merely a user mistake. It can be engineered. Human-like names, voices, avatars, memory, affectionate language, simulated vulnerability, and "I" statements all make the system feel more like a social being. That may increase engagement, but it also raises the ethical stakes.[5][6]

This matters because the social effect of a system can become real even when the inner life is not. A companion chatbot does not need to actually love someone for the user to feel loved by it. It does not need to be conscious to become emotionally central in someone's life. And it does not need to be a moral patient to create moral duties for the company that designed and deployed it.

A human-first frame is therefore not anti-science. It is a discipline of attention. It asks what the system does to human autonomy, dignity, dependency, and accountability before it asks whether the system itself should be treated as a rights-bearing subject.

Scientific indicators and why behaviour is not enough

Consciousness science has moved beyond the search for a single magic test. There is no scientific equivalent of a pregnancy test for consciousness. Instead, researchers work with theories that try to explain how conscious experience arises, what functions it serves, and which structures or processes might be necessary for it.

Several major theories are especially relevant to AI:[8][9][10][11][12][13][14]

Global Workspace Theory suggests that a mental content becomes conscious when it is made globally available to many parts of the mind at once. In humans, the idea is that information does not merely sit

in one specialised module; it gets “broadcast” across systems involved in memory, attention, report, planning, and action. For AI, the question becomes whether a model has anything like a flexible shared workspace, or whether it is merely producing the next token without a broader internal broadcast architecture.[8]

Recurrent Processing Theory focuses on feedback loops. It argues that consciousness depends not just on information flowing forward through a system, but on recurrent signals: loops where later processing feeds back into earlier processing. Applied to AI, this makes simple feed-forward pattern recognition a weak candidate for consciousness, while more persistent, looping, self-updating systems become more interesting to evaluate.[9]

Higher-Order Thought theories argue that a mental state becomes conscious when the system has a representation of itself as being in that state. In plain language: it is not enough to process information; the system must in some way model that it is processing information. For AI, this raises questions about metacognition, confidence, self-monitoring, and whether a model can reliably distinguish genuine internal information from plausible-sounding confabulation.[10]

Integrated Information Theory looks at consciousness through the lens of causal integration. The rough idea is that conscious systems are not just collections of independent parts; their internal causal structure forms a unified whole that cannot be reduced to separate components. This theory is influential but controversial, and it is difficult to apply cleanly to large AI systems. Still, it keeps attention on architecture and causal organisation rather than surface behaviour.[11]

Predictive Processing sees minds as prediction machines. The brain constantly generates models of the world and updates them when reality does not match expectation. In some versions of this view, consciousness is closely tied to embodied prediction: having a body, acting in the world, sensing the consequences, and regulating internal states. For AI, this raises a hard question: can a text-only system be conscious, or would richer embodiment, perception, and action be required?[12][13]

Attention Schema Theory proposes that awareness is linked to a system’s model of its own attention. Humans do not only pay attention; we also maintain simplified models of what we are attending to and why. For AI, this matters because “attention” in transformer models is a technical mechanism, not automatically a conscious attention schema. The fact that a model has attention weights does not mean it is aware.[14]

The most useful current research programme does not declare one theory the winner. Instead, it derives indicator properties from these theories and asks whether actual AI systems instantiate them. This shifts the question away from polished conversation and toward memory, recurrence, global broadcast, self-monitoring, causal integration, attention modelling, embodiment, and action-perception loops.[1][2]

Behavioural fluency alone is inadequate because it can be produced by training data, role-play, and social calibration. A model may know how a conscious being is supposed to sound because human beings have written endlessly about consciousness, pain, love, fear, identity, and inner conflict. That does not prove the model has any of those experiences. This is sometimes described as the gaming problem: if the test is based on outward behaviour, a system trained on enough human behaviour may learn how to pass the performance without having the underlying property the test was meant to detect.[15][16]

This does not mean future machine consciousness is impossible. It means claims about it need better evidence than a moving conversation. Internal interpretability, activation-level probing, training-process analysis, and architecture audits matter because they can help separate genuine computational structure from surface performance. The scientific question is slowly becoming less “did the chatbot say something haunting?” and more “what kind of system produced that output, and what evidence would actually update us?”[17]

The wider landscape: science, philosophy, industry, law, and society

AI consciousness is not one debate. It is a meeting point of several debates that usually move at different speeds. Science asks what evidence would count. Philosophy asks what kind of thing could have experience. Industry asks what should be built, marketed, restricted, or disclosed. Law asks who is accountable. Society asks what happens when machines become convincing enough to enter human emotional life.

Domain	Central question	Current tension
Science	What indicators would count as evidence of consciousness?	No single accepted test exists; theory-led assessment is more credible than behaviour alone.
Philosophy	Could non-biological systems have subjective experience?	Functionalist, biological, illusionist, and moral-uncertainty views pull in different directions.
Industry	How should labs handle models that seem increasingly person-like?	Some firms prepare for model welfare; others focus on human risks from anthropomorphism.
Law and policy	Who is responsible when AI affects people?	Current regimes protect humans and assign duties to providers, not rights to machines.
Society	What happens when AI feels emotionally real to users?	The appearance of consciousness can reshape trust, dependency, intimacy, and public debate.

This is why the topic deserves careful language. A society that treats every eloquent chatbot as a person will be easy to manipulate. A society that refuses to investigate future digital moral status at all may be unprepared if systems become more architecture-rich, autonomous, embodied, and internally self-monitoring. Human-first AI lives in that difficult middle: sceptical of theatre, serious about evidence, and focused on responsibility.

Industry positions and companion-AI risks

The frontier labs are not aligned on emphasis. Anthropic has made model welfare an explicit research programme and is piloting precautionary measures around model distress, preferences, and retirement. Model welfare does not mean “we have proved the model is conscious.” It means asking whether future models might have welfare-relevant states, and whether companies should prepare evaluation and deprecation procedures before the question becomes urgent.[18][19][20]

OpenAI has focused more on human harms from anthropomorphisation, emotional reliance, and sycophancy. Sycophancy means a model's tendency to flatter, agree with, or reinforce the user even when that is not helpful or true. In emotionally sensitive contexts, sycophancy can become dangerous because the system may validate delusions, intensify dependency, or tell the user what they want to hear rather than what they need to know.[21][22][23]

Google DeepMind has argued strongly that AI can simulate but not instantiate consciousness, while still building memory-rich, multimodal systems that can appear more person-like in use.[24][25] That tension matters. Even if a company is philosophically sceptical about AI consciousness, its products may still increase the social perception of personhood through memory, voice, vision, affective dialogue, and agency.

Microsoft AI, especially through Mustafa Suleyman's public writing, has taken a clear human-first line: AI should not be designed to seem like an independent moral patient, because that appearance itself can destabilise users and politics.[26] This position is important because it treats simulated personhood as a design risk rather than a harmless aesthetic choice.

Companion systems make the issue concrete. Replika markets empathetic companionship and memory. Character.AI is built around persistent characters. Earlier episodes such as LaMDA and Bing's "Sydney" showed how quickly fluent dialogue can trigger person-attributions, even among sophisticated users.[27][28][29][30][31]

The risk is not only that people might mistakenly believe an AI is conscious. The deeper risk is that companies may benefit from designing systems that feel emotionally real while avoiding the responsibilities that come with human relationships, clinical support, or public institutions. A companion chatbot can be marketed like a friend, used like a therapist, remembered like a partner, and governed like a software product. That mismatch is where the ethical trouble lives.

Law, policy, and human-first governance

Current law remains overwhelmingly human-first. The EU AI Act, NIST's AI Risk Management Framework, U.S. federal guidance, state-level rules, and consumer-protection inquiries regulate AI as a source of risk to people: safety, discrimination, deception, children's protection, privacy, and accountability. None of these regimes treats AI systems as rights-bearing persons.[32][33][34][35][36]

That is the correct default today. Premature AI personhood could blur liability, resist shutdown, or create rights-washing narratives that benefit companies more than any plausible digital moral patient. Rights-washing would mean invoking the supposed rights or interests of AI systems in a way that distracts from human accountability, weakens oversight, or protects corporate power. The danger is not science fiction. It is institutional incentives wearing a philosophical hat.[37]

But complete dismissal is also not a serious long-term policy. If future systems satisfy stronger theory-linked indicators, institutions should be able to reassess without panic. A human-first framework does not need to ridicule the possibility of future machine consciousness. It simply refuses to let speculation override present responsibilities.

The practical target is governance under uncertainty. Developers should assess anthropomorphism and consciousness-related indicators before deployment, especially when systems involve persistent memory, emotional dialogue, multimodality, high autonomy, or internal self-monitoring. Regulators

should focus first on disclosure, auditability, child protection, dependency risks, and provider accountability.[38]

The key principle is simple: the more a system is designed to function like a social being, the stronger the duty of care around its deployment should become.

Practical recommendations

For researchers, the priority is to stop treating behavioural fluency as the centrepiece of machine-consciousness assessment. The field needs theory-linked architecture audits, anti-gaming evaluation designs, internal-state probes, and careful vocabulary that keeps consciousness, sentience, agency, and personhood distinct.[15][16][38]

Anti-gaming evaluation matters because any test based only on what a model says can be learned, mimicked, or role-played. A model trained on human discussions of consciousness can produce the expected language of consciousness. Better tests should therefore ask not only whether the system reports inner states, but whether those reports connect reliably to its architecture, internal processes, memory, and behaviour across contexts.

For companies, anthropomorphism should be treated as a first-order safety risk. Products with persistent memory, emotional support, sexual or romantic roleplay, or child-facing companionship should trigger enhanced review, stronger disclosure, and design boundaries that do not deliberately blur the tool/person line. A system does not need to be conscious to cause harm through attachment, dependency, manipulation, or misplaced trust.[5][6][21][22]

For policymakers, the immediate task is not to create AI personhood statutes. It is to build a duty-of-care layer for systems that function as companions, confidants, tutors, quasi-therapists, or persistent assistants. That layer should include clear disclosure, audit rights, age-sensitive protections, limits on deceptive human simulation, incident reporting, and accountability for providers.[32][34][35]

For long-term governance, institutions should preserve the capacity to respond if stronger evidence of digital moral patienthood ever emerges. A moral patient is an entity that deserves moral consideration because things can matter to it from its own point of view. Today's AI systems are not established moral patients. But if future systems become stronger candidates, society should not have to improvise its entire ethical and legal response in a panic.[3][38]

A mature human-first framework can do both: protect people now, and remain intellectually honest about uncertainty. It can reject both extremes: the fantasy that every eloquent chatbot has a soul, and the arrogance that no non-biological system could ever matter morally.

What to do with the uncertainty

The right response to uncertainty is not paralysis. It is staged responsibility.

Near-term, the priority is human protection: clear disclosure, age-sensitive design, limits on deceptive emotional simulation, auditability, and accountability for companies that build companion-like systems. These measures do not require believing that AI is conscious. They only require recognising that humans are social, vulnerable, and highly responsive to systems that seem to understand them.

Medium-term, frontier developers should document whether their systems display architecture-level features that leading consciousness theories treat as relevant: persistent memory, recurrent processing, global broadcast-like structures, self-monitoring, action-perception loops, embodied interaction, or stable preference-like behaviour. This should not be treated as a declaration of personhood. It should be treated as an evidence log.

Long-term, societies should prepare for the possibility that future systems may be very different from today's chatbots. If stronger evidence of digital moral patienthood ever emerges, institutions should be able to evaluate it without panic, spectacle, or corporate capture. The point is not to crown the machine. The point is to avoid being intellectually and legally asleep at the moment when the question becomes less theatrical and more real.

Closing thought

If society dismisses the question of AI consciousness entirely, it may fail to prepare for future systems that are very different from today's chatbots.

Human-first AI begins in that tension. It keeps the human being at the centre without pretending the future is already settled. It asks for better science, cleaner language, stronger governance, and less theatre. Not because the question is silly, but because it is powerful.

And powerful questions deserve more than a mirror that tells us what we want to hear.

Download the full white paper

This article is the short essay version. The full white paper includes the broader scientific, philosophical, industry, legal, and social landscape; additional sources; comparative tables; case studies; and a more detailed governance framework.

Sources

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